

12 - Nitrogen & Sulfur

22/12/24

Reactivity of nitrogen

- unreactive due to its strong triple bond ($N \equiv N$) which require a lot of energy to break & because it is non-polar (both atoms have the same electronegativity), reducing its tendency to react with other substances

Ammonia

⇒ basicity of ammonia

- acts as a base because it can accept a proton $[\therefore NH_3(aq) + H^+(aq) \rightarrow NH_4^+(aq)]$

⇒ ammonium ion

- a tetrahedral molecule with all N-H bonds equivalent

- formed when ammonia reacts with an acid $[\therefore NH_3(aq) + HCl(aq) \rightarrow NH_4Cl(aq)]$

⇒ formation of ammonia by displacement



* the OH^- ion removes a proton (H^+) from NH_4Cl to form ammonia gas

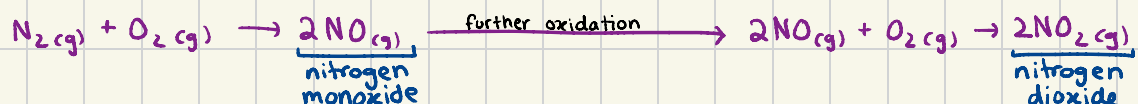
Formation of oxides of nitrogen

→ nitrogen monoxide/dioxide are toxic & harmful gases

i) natural

→ lightning

high energy lightning can break the $N \equiv N$ bond, enabling it to react with oxygen (oxidise)



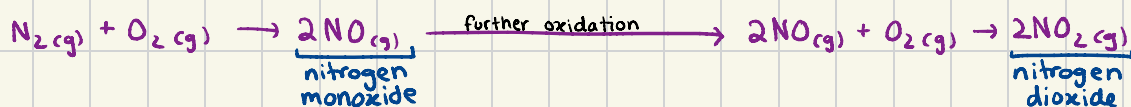
→ soil microbial activity

certain soil bacteria release NO & NO_2 during their metabolic processes (denitrification)

ii) man-made

→ combustion in engines

at high temp. in car engines, nitrogen gas reacts with oxygen to form oxides of nitrogen



* car engines also release carbon monoxide via incomplete combustion between carbon & oxygen and also volatile organic compounds (VOCs) which are usually unburnt hydrocarbons

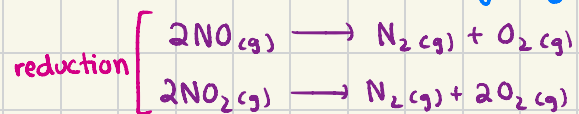
→ industrial processes

power plants & factories burning fossil fuels release NO and NO_2

Removal of oxides of nitrogen

* catalytic converters are used in cars to reduce these toxic gases

1) reduce NO & NO₂ to nitrogen gas

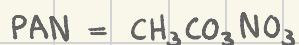


2) reaction of the oxides with carbon monoxide



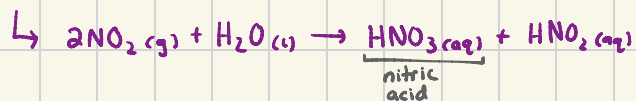
Formation of PAN (peroxyacetyl nitrate)

- NO & NO₂ can react with ^{VOCs}unburned hydrocarbons in sunlight to form PAN which is a photochemical smog



Acid Rain

- HNO₃ (nitric acid) is a major contributor to acid rain because it dissolves in rainwater, decreasing the pH and making it acidic
- when NO or NO₂ react with water vapor, nitric acid is produced



- H₂SO₄ (sulfuric acid) is another major contributor to acid rain
- NO₂ can act as a catalyst to convert SO₂ → SO₃ which reacts with water to make H₂SO₄

